# Adrian Maurice MÜLTHALER

muelthaler.com nairdanus

## Education

10.2024 – 09.2026 **M.Sc., Ludwig-Maximilian-University of Munich**
in Computational Linguistics and Computer Science.
Current grade: 1.04

10.2021 – 09.2024 **B.Sc., Ludwig-Maximilian-University of Munich**
in Computational Linguistics and Computer Science.
Overall grade: 1.13 (GPA 95.67%)
*Simulating Circuits in Quantum Natural Language Processing using Tensor Networks*

07.2021 **European Baccalaureate, European School Munich**
Grade: 93.54%

## Employment History

10.2025 – 02.2026 **Tutor for Deep Learning for Natural Language Processing,**
Center for Information and Language Processing, LMU Munich
- Deep Learning Basics
- Transformer Architecture
- Large Language Models
- Decoding Strategies
- Reinforcement Learning from Human Feedback

since 03.2025 **Student Research Assistant for MUNIQC-Atoms,**
Munich Network Management Research Group, LMU Munich
Developing a Visual Programming Interface for MBQC
- Building a C++ backend server using Crow
- Building a TypeScript frontend server using React

09.2024 – 02.2025 **Student Research Assistant,**
Center for Information and Language Processing, LMU Munich
Collaborative Research Center for Expert-AI Cooperation
Accreditation of Computational Linguistics Study Program
- Formatting and Organizing Complex Documents in LaTeX

04.2024 – 08.2024 **Tutor for Introduction to Quantum Computing,**
Institute for Informatics, LMU Munich
- Introduction to Quantum Mechanics
- Quantum Bits / Registers
- Entanglement, Quantum Teleportation, Dense Coding
- Quantum Algorithms e.g. Grover's, Shor's, VQA
- Quantum Cryptography

## Employment History (continued)

**Tutor for Mathematical Methods in Computational Linguistics,**
Center for Information and Language Processing, LMU Munich
- Discrete Mathematics
- Linear Algebra
- Stochastics

10.2022 – 02.2025 **Tutor for Introduction to Computational Linguistics,**
Center for Information and Language Processing, LMU Munich
- Introduction to Linguistics
- Automata Theory / Context-Free Grammars
- Statistical Language Models e.g. Naive Bayes, Hidden Markov Models

11.2021 – 09.2023 **Working Student in the field of Data Science,**
Knowlix GmbH
- Data Generation with Jinja2 and HTML/CSS templates
- Web Scraping using Python

## Skills

**Coding**
- Python
- C++
- Java
- Haskell

**Web Dev**
- HTML
- CSS
- JavaScript
- LAMP Web Server

**Languages**
German — Native Speaker
Romanian — Native Speaker
English — Proficient (IELTS: Band 8)
Spanish — Basic Proficiency
French — Limited Proficiency ($A2+$)
Chinese — Limited Proficiency ($HSK2$)

## Research Publications

S. Loftus, **A. Mülthaler**, S. Hoeken, S. Zarrieß, and O. Alacam, "Using LLMs and preference optimization for agreement-aware HateWiC classification," in *Proceedings of the The 9th Workshop on Online Abuse and Harms (WOAH)*, Vienna, Austria: Association for Computational Linguistics, Aug. 2025, pp. 538–547, ISBN: 979-8-89176-105-6. URL: https://aclanthology.org/2025.woah-1.47/.

## Extracurricular Activities

Student Council for Computational Linguistics

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|---|---|
| 10.2025 – 10.2026 | Chair |
| 10.2024 – 10.2025 | Representative at the Faculty Committee for Study Grants |
| 10.2024 – 10.2025 | Elected Representative in the Faculty Council |
| 10.2024 – 10.2025 | Delegate to the Student Representative Council |

Posters and Presentations

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|---|---|---|
| 08.2025 | - | Poster Presentation at *Association for Computational Linguistics, (ACL)*, Vienna |
| 05.2025 | - | Oral Presentation at *Tagung der Computerlinguistik Studierenden (TaCoS)*, Bochum |

Interests
Reasoning Models - Reinforcement Learning - Agentic AI Systems - Quantum Computing